

Research Paper

Product Optimization & Revenue Contribution Analysis

Afficionado Coffee Roasters

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Tools Used: Python, Pandas, Matplotlib, Seaborn, Plotly, Streamlit

1. Abstract

This research paper presents a data-driven analysis of product performance and revenue contribution at Afficionado Coffee Roasters. Using transaction-level data of 1,49,116 records across 3 store locations, this study identifies top and bottom performing products, measures revenue concentration, and provides actionable recommendations for menu optimization.

The analysis reveals that Coffee dominates revenue at 38.63%, while 42 out of 80 products contribute 80% of total revenue — indicating an opportunity to streamline the menu and improve operational efficiency.

2. Introduction

2.1 Background

In specialty coffee retail, not all products contribute equally to revenue. Afficionado Coffee Roasters operates 3 store locations — Lower Manhattan, Hell's Kitchen, and Astoria — with a menu of 80 products across 9 categories.

Despite having rich transaction data, the business lacked:

- Clear visibility into product popularity vs profitability
- Category-level revenue dependency insights
- Identification of low-impact menu items

2.2 Problem Statement

Menu decisions were being made without structured product analytics, leading to:

- Overcrowded menu slowing service
- Resources spent on low-performing products
- Missed revenue opportunities from hero products

2.3 Research Objectives

Primary:

- Identify top-selling and least-selling products
- Quantify revenue contribution by product and category
- Measure revenue concentration across the menu

Secondary:

- Identify high-impact "Hero" products
- Highlight low-performing products for review
- Support menu simplification strategy

3. Dataset Description

Attribute	Details
Total Records	1,49,116 transactions
Time Period	Year 2025
Store Locations	3 (Lower Manhattan, Hell's Kitchen, Astoria)
Total Products	80 unique products
Product Categories	9 categories
Missing Values	0 (clean dataset)

3.1 Key Columns Used

- **transaction_qty** — Units purchased per transaction
- **unit_price** — Price per unit
- **product_category** — Broad product group
- **product_type** — Product variant
- **product_detail** — Exact product name
- **store_location** — Branch name

3.2 Derived Column

`Revenue = transaction_qty × unit_price`

This column was computed for all 1,49,116 records.

4. Methodology

Step 1 — Data Ingestion & Validation

- Loaded Excel dataset using Pandas
- Checked for missing values → Zero found
- Validated quantities (min = 1, no negatives)
- Validated prices (min = \$0.80, max = \$45.00)

Step 2 — Revenue Computation

- Calculated revenue at transaction level
- Aggregated by product, product type, and category

Step 3 — Product Popularity Analysis

- Counted total units sold per product
- Ranked products by sales volume
- Identified top 10 and bottom 10 performers

Step 4 — Revenue Contribution Analysis

- Calculated total revenue per product
- Computed revenue share percentage
- Compared volume rank vs revenue rank

Step 5 — Category & Product Type Analysis

- Aggregated revenue by category
- Analyzed product type contribution within categories

Step 6 — Pareto Analysis (80/20 Rule)

- Calculated cumulative revenue percentage
- Identified products contributing 80% of revenue

- Classified revenue anchors vs long-tail products

Step 7 — KPI Calculation

- Product Revenue Contribution %
- Product Sales Volume
- Category Revenue Share
- Revenue Concentration Ratio
- Product Efficiency Score (Revenue per Unit)

Step 8 — Hero Product Identification

- Normalized volume, revenue, and efficiency scores
- Calculated composite Hero Score
- Ranked products by Hero Score

5. Analysis & Findings

5.1 Overall Revenue Summary

KPI	Value
Total Revenue	\$6,98,812.33
Total Units Sold	2,14,470
Total Transactions	1,49,116
Average Revenue per Transaction	\$4.69
Highest Single Transaction	\$360.00

5.2 Product Popularity Analysis

Top 10 Best Selling Products:

Rank	Product	Units Sold
1	Earl Grey Rg	4,708
2	Dark Chocolate Lg	4,668
3	Morning Sunrise Chai Rg	4,643

Rank	Product	Units Sold
4	Latte	4,602
5	Peppermint Rg	4,564

Bottom 5 Products:

Rank	Product	Units Sold
76	Jamaican Coffee River	146
77	Earl Grey	142
78	Guatemalan Sustainably Grown	134
79	Spicy Eye Opener Chai	122
80	Dark Chocolate	118

Key Finding: Regular (Rg) and Large (Lg) sizes dominate top sellers. Small and no-size variants consistently underperform.

5.3 Revenue Contribution Analysis

Top 5 Products by Revenue:

Product	Revenue	Share %
Sustainably Grown Organic Lg	\$21,151	3.03%
Dark Chocolate Lg	\$21,006	3.01%
Latte Rg	\$19,112	2.73%
Cappuccino Lg	\$17,641	2.52%
Morning Sunrise Chai Lg	\$17,384	2.49%

Key Finding: Volume rank $\neq$ Revenue rank. Earl Grey Rg is #1 in volume but only #26 in revenue. Sustainably Grown Organic Lg is #1 in revenue despite lower volume — due to higher unit price.

5.4 Category Revenue Analysis

Category	Revenue	Share %
Coffee	\$2,69,952	38.63%

Category	Revenue	Share %
Tea	\$1,96,405	28.11%
Bakery	\$82,315	11.78%
Drinking Chocolate	\$72,416	10.36%
Coffee Beans	\$40,085	5.74%
Branded	\$13,607	1.95%
Loose Tea	\$11,213	1.60%
Flavours	\$8,408	1.20%
Packaged Chocolate	\$4,407	0.63%

Key Finding: Coffee + Tea = 66.74% of total revenue. Top 4 categories = 88.88% of revenue. Business is heavily dependent on Coffee and Tea.

5.5 Product Type Analysis

Coffee Category:

- Barista Espresso = 33.86% of coffee revenue
- Gourmet Brewed = 25.94%

Tea Category:

- Brewed Chai Tea = 39.25% of tea revenue

Bakery Category:

- Scones = 44.79% of bakery revenue

Key Finding: Drinking Chocolate has only 1 product type (Hot Chocolate = 100%). There is an opportunity to add variety.

5.6 Pareto Analysis

Metric	Value
Total Products	80
Products for 80% Revenue	42

Metric	Value
Menu % needed	52.5%
Long-tail products	38 (47.5%)
Long-tail revenue	~20%

Key Finding: This is not a classic 80/20 result. Revenue is well-distributed across products, indicating a healthy menu balance. However, 38 long-tail products still need review.

5.7 Hero Products

Rank	Product	Hero Score	Revenue
1	Dark Chocolate Lg	69.5	\$21,006
2	Sustainably Grown Organic Lg	68.4	\$21,151
3	Latte Rg	65.1	\$19,112
4	Latte	62.6	\$17,257
5	Morning Sunrise Chai Lg	61.1	\$17,384

6. Key Insights

1. **Coffee is King** — 38.63% revenue share. Any disruption to coffee supply or quality will heavily impact business.
2. **Size Matters** — Regular (Rg) and Large (Lg) sizes dominate both volume and revenue. Small sizes underperform consistently.
3. **Volume ≠ Revenue** — Earl Grey Rg sells most units but ranks 26th in revenue. Price matters as much as popularity.
4. **Balanced Revenue** — 52.5% products drive 80% revenue (not classic 80/20). This shows good menu diversity.
5. **Dark Chocolate Lg is the True Hero** — Balanced across volume, revenue, and efficiency.
6. **Chocolate Variety Gap** — Drinking Chocolate has only 1 product type. New flavors could unlock additional revenue.

7. Recommendations

Immediate Actions

Action	Detail
Promote Hero Products	Dark Chocolate Lg, Sustainably Grown Organic Lg, Latte Rg
Review Bottom 10	Consider removing or repricing Dark Chocolate (no size), Spicy Eye Opener Chai
Push Large Sizes	Customers prefer Lg/Rg — promote these sizes
Add Chocolate Variety	Only 1 type exists — new flavors can grow revenue

Strategic Actions

Action	Detail
Protect Coffee & Tea	66.74% revenue — maintain quality and supply
Simplify Menu	80 items is too many — consider reducing to 60
Price Optimization	Low-efficiency products may need price revision
Focus on Barista Espresso	33.86% of coffee revenue — train staff, promote drinks

8. Conclusion

This analysis successfully shifts focus from operational data to actionable product intelligence. By analyzing 1,49,116 transactions across 80 products and 3 stores, we identified clear revenue drivers, underperforming products, and strategic opportunities.

The key takeaway is that **data-driven menu decisions can directly improve revenue and operational efficiency** at Afficionado Coffee Roasters.

9. References

- Dataset: Afficionado Coffee Roasters Transaction Data (2025)
- Tools: Python 3.14, Pandas, Matplotlib, Seaborn, Plotly, Streamlit
- Internship: Unified Mentor Data Analyst Program

- Live Dashboard: <https://afficionado-coffee-analysis-ip2nceibngh4brxb4av5iw.streamlit.app/>

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